Deploying Web App on ECS - Step 9: Verification of Services

To Begin with the Lab

Summary of the Lab

In this final step, the running **Amazon ECS** service was verified end to end after auto scaling had already been enabled. The application was opened through the **Application Load Balancer** DNS name, confirming that the service was publicly reachable through the load balancer. The ECS tasks were checked to make sure they were running and healthy, and the target group was reviewed to confirm that both tasks were registered and healthy behind the ALB. The application was also tested by uploading a file, which was successfully stored in **Amazon S3**. After verification, the lab explained the cleanup process for deleting the scheduled scaling actions, service, cluster, NAT Gateway, VPC, and Elastic IP to avoid extra charges.

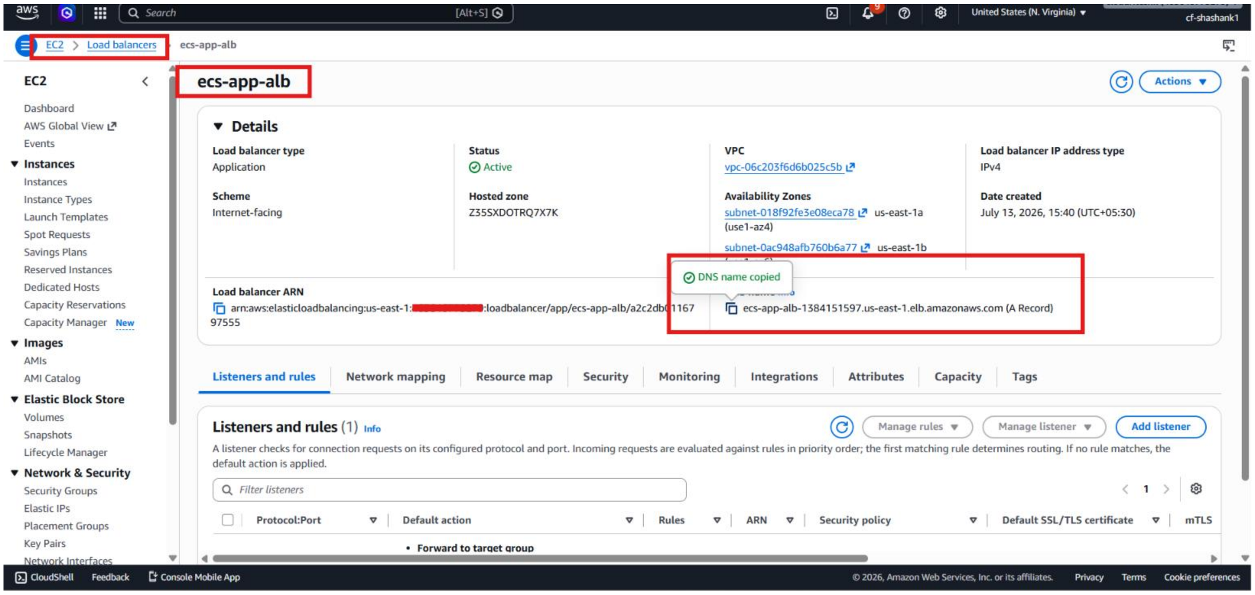
Understanding Service Verification in ECS

Service verification in **Amazon ECS** means checking that the full application path works correctly from the user request to the running container and back. It exists so you can confirm that the load balancer, ECS tasks, target group, and storage integration are all working together. If any part fails, the app may still exist but users may see errors such as 502, unhealthy targets, or upload failures. The benefit is that you can catch networking, IAM, or storage issues before handing the system over to real users.
Example:
A user opens the app through the **Application Load Balancer** DNS name and uploads my_aws.png. If the ECS service is healthy, the task receives the request, the file is saved in **Amazon S3**, and the target group shows both tasks as healthy. If the tasks are not healthy or the bucket is wrong, the upload or page load will fail.

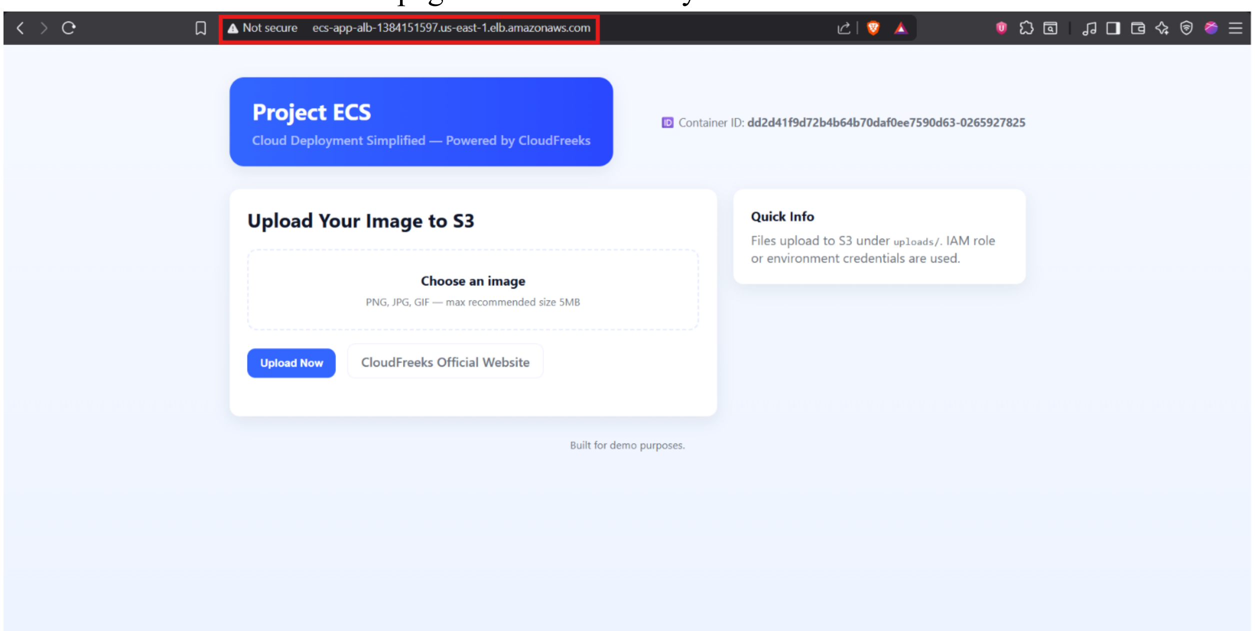
Lab Steps

Step 1 – Open the application through the ALB

- Sign in to the **AWS Management Console**.
- Open the saved **Application Load Balancer** DNS name in a browser.

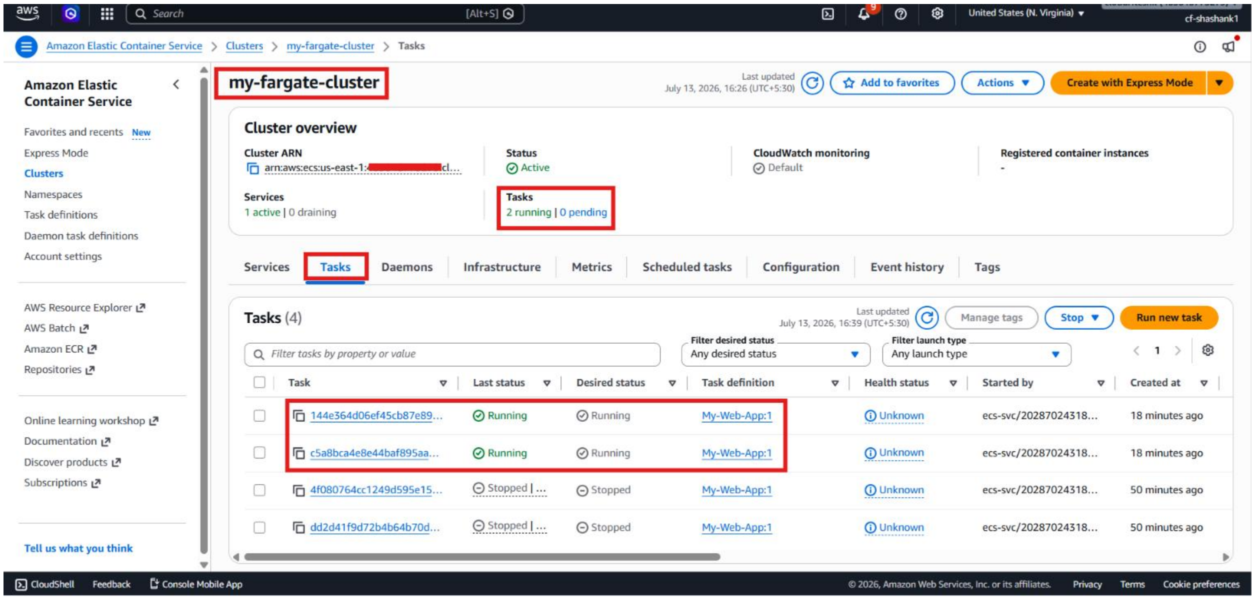


- Confirm that the PHP web page loads successfully.



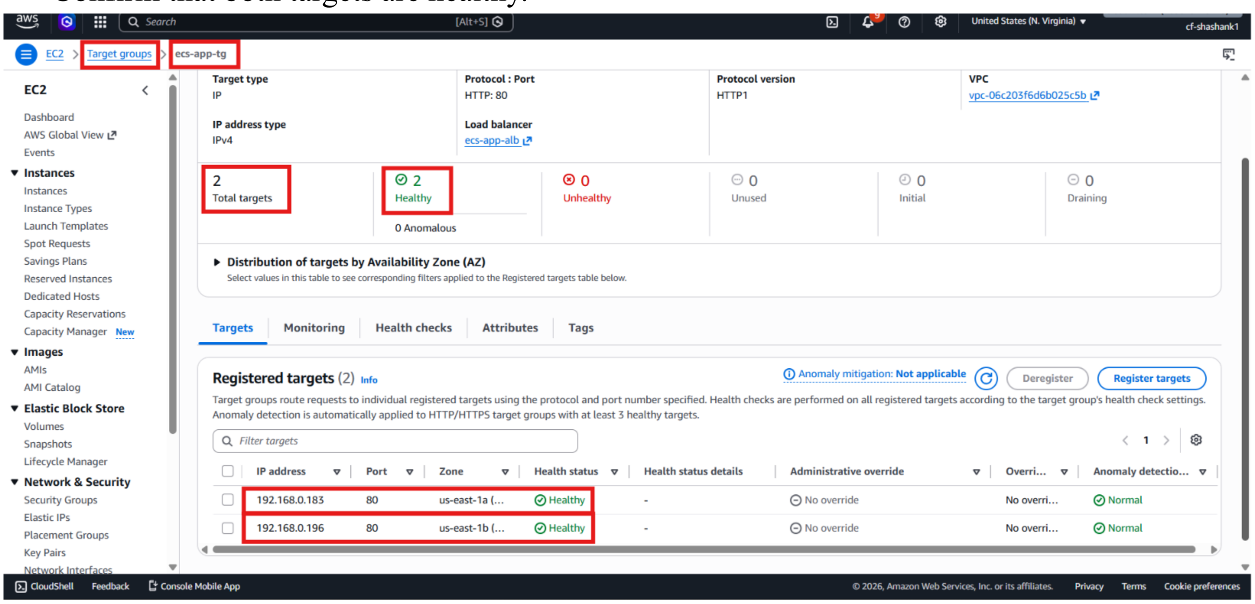
Step 2 – Verify ECS tasks

- Open **Amazon ECS**.
- Open the cluster and service.
- Confirm that the desired number of tasks is running.
- Open the task details and verify that the tasks are in the running state.



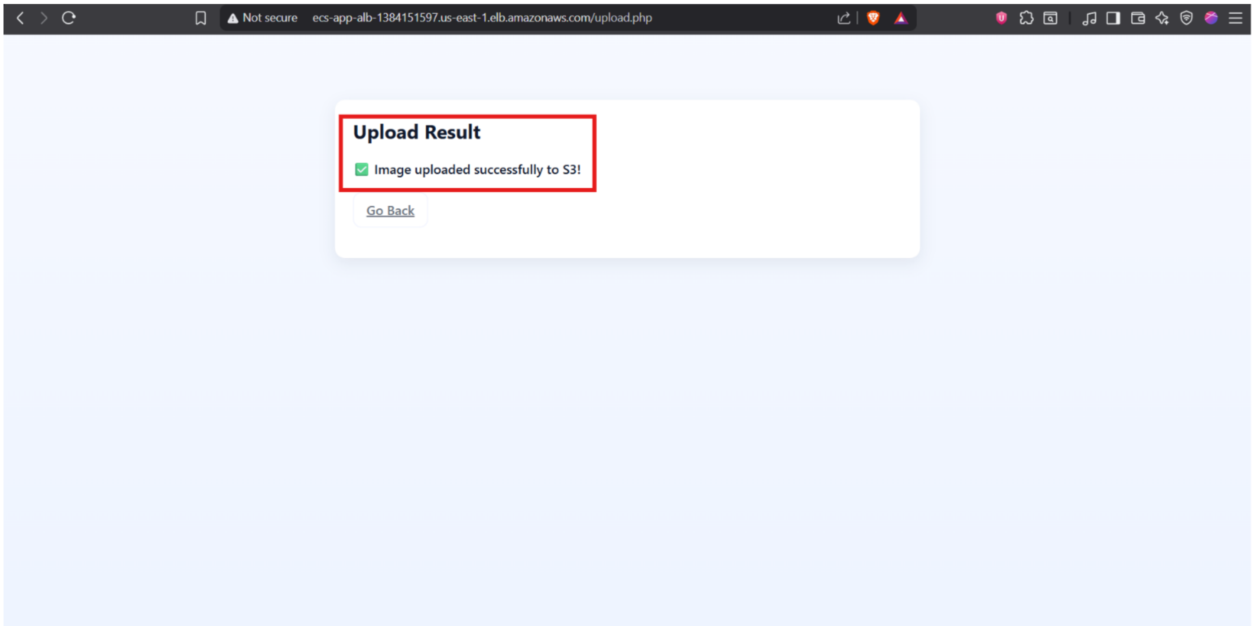
Step 3 – Verify the target group

- Open **EC2**.
- Go to the target group used by the load balancer.
- Confirm that both ECS tasks are registered.
- Confirm that both targets are healthy.

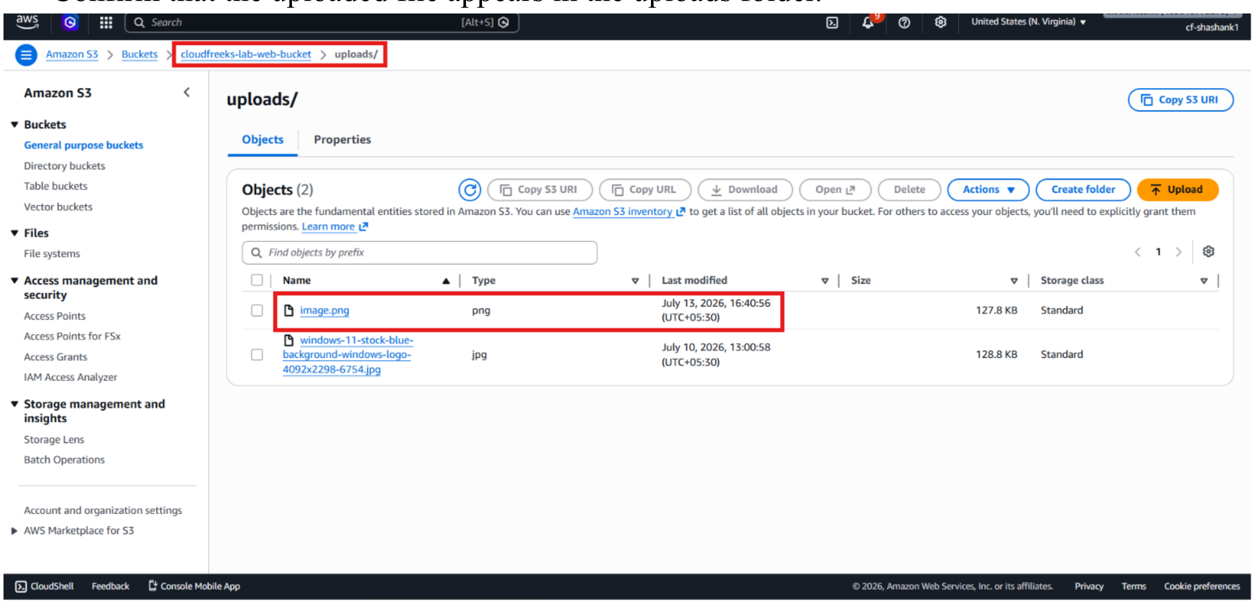


Step 4 – Test file upload

- Open the application page.
- Select a file from the local system.
- Click **Upload**.
- Confirm that the upload succeeds.

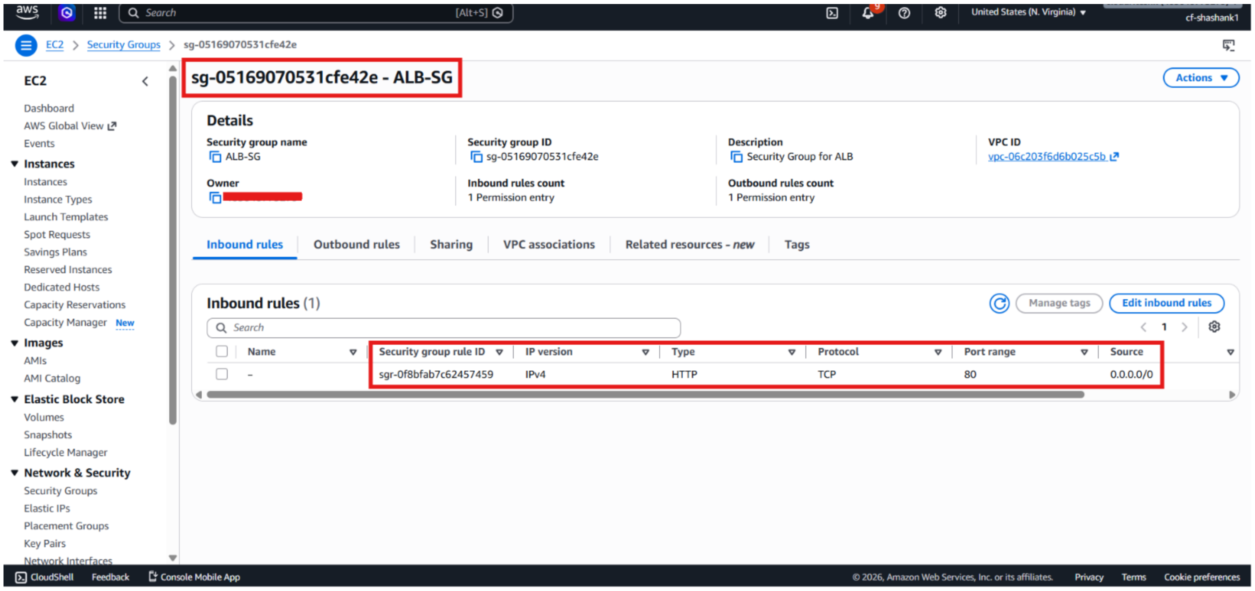


- Open **Amazon S3**.
- Refresh the bucket.
- Confirm that the uploaded file appears in the uploads folder.

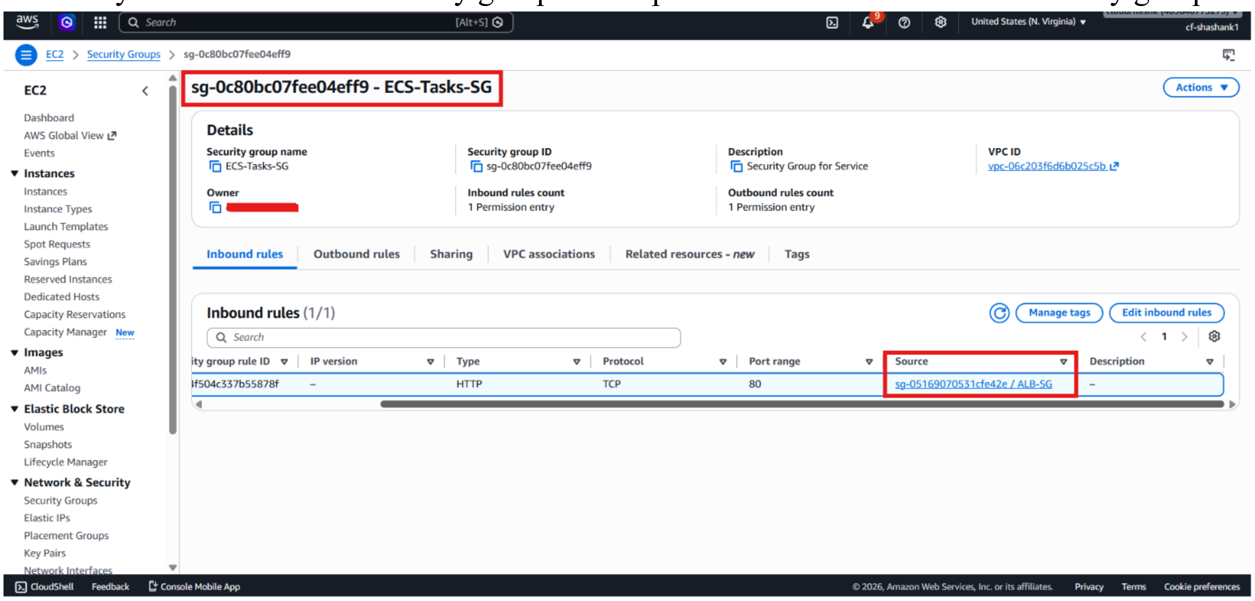


Step 5 – Check common issues if the app does not open

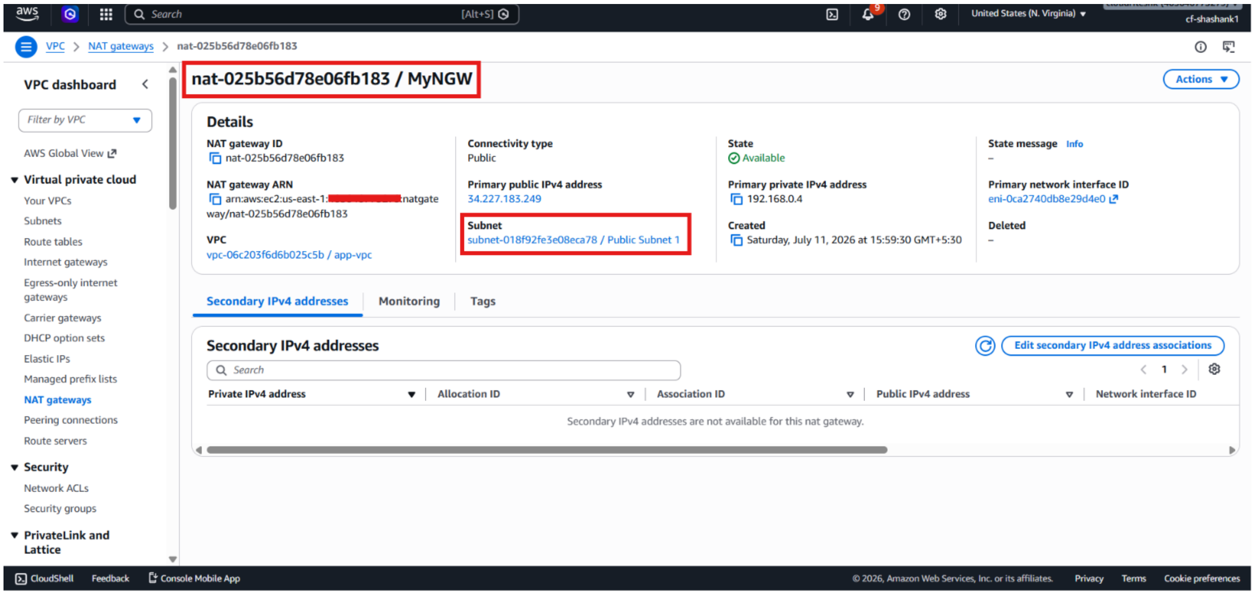
- Verify that the load balancer security group allows **HTTP** traffic from 0.0.0.0/0.



- Verify that the ECS task security group allows port 80 traffic from the ALB security group.



- Verify that the task is in the running state.
- Verify that the **NAT Gateway** is in the public subnet and the private route table points to it.



- Verify that the S3 bucket name and Region in the task definition are correct.

Step 6 – Clean up the lab resources

- Open the ECS service auto scaling settings.
- Delete the scheduled scale-out and scale-in actions.
- Disable auto scaling if needed.
- Delete the ECS service with force delete if required.
- Delete the ECS cluster.
- Delete the NAT Gateway.
- Delete the **Elastic IP**.
- Delete the **VPC** only after all subnets and gateways are removed.

Verification

- Verified that the application opened successfully through the **Application Load Balancer** DNS name.
- Verified that the ECS service was running with the expected number of tasks.
- Verified that both ECS tasks were healthy in the target group.
- Verified that the application upload worked correctly.
- Verified that the uploaded file was stored in **Amazon S3**.
- Verified that the load balancer, ECS service, and target group were connected properly.
- Verified that the cleanup steps were identified to avoid extra AWS charges.